

Fountain Head Hybrid Energy Financeability Review

Owner-facing preliminary screening report

Why a 60% renewable hybrid case appears more financeable than both the short-life diesel replacement case and a 100% renewable screen

Prepared for	Early discussion with Fountain Head mine owner / project sponsor
Prepared as	Independent early-stage screening note
Model basis	Public-data desktop reconstruction plus indicative hybrid energy financeability model
Date	10 June 2026

Positioning statement

This report is designed to support an informed first conversation. It is not an EPC design, bankable feasibility study, financing offer, legal opinion or investment recommendation.

Executive Summary

Core message. The Fountain Head energy question should not be framed as a simple choice between diesel and 100% renewable power. The more useful question is which hybrid energy structure can lower the owner's energy cost, reduce diesel exposure, remain reliable for mine operations, and be financeable under a third-party SPV / energy-as-a-service structure.

The screening conclusion is clear: the 60% renewable hybrid case is currently the most balanced option. It passes simplified lender DSCR screening and produces positive customer true saving after the required buy-out. The 100% renewable case can be re-gearred to pass a 1.30x DSCR, but it fails customer economics because the required buy-out is too large. The short-life standalone case is not a strong SPV candidate because the initial mine-life window is too short to recover the full system cost.

Area	Owner-facing finding
Short-life standalone case	Not suitable for a third-party SPV structure. The project life is too short and the required buy-out is too high.
Expanded hub / longer-life case	A 7-year energy demand horizon materially improves financeability, reflecting the post-merger regional asset context.
Best current screen	The 60% renewable hybrid case passes DSCR and produces positive true customer saving after buy-out.
100% renewable screen	Useful as an upper-bound decarbonisation case, but not commercially preferred under current assumptions.

Table 1. Executive findings from the current screening model.

Important disclaimer

This report is an independent preliminary screening review prepared using publicly available information, indicative assumptions and high-level financial modelling. It supports early-stage discussion only. It is not an EPC design, feasibility study, investment recommendation, financing offer, legal advice, tax advice or technical due diligence report. All outputs must be validated against owner-provided mine plans, load data, diesel consumption records, hourly dispatch modelling, vendor quotations, permitting requirements and lender requirements before any commercial decision is made.

Recommended decision pathway

Use S4, not S5, as the next engineering and commercial base case. The next decision should not be whether to pursue 100% renewable power immediately. The better pathway is to validate the 60% renewable case through hourly dispatch modelling, then use that validated design to negotiate tariff structure, diesel cost allocation, buy-out mechanism, carbon benefit treatment and ownership options.

1. Project context: why the energy structure matters

Fountain Head is not a normal grid-connected energy cost exercise. It is a mining restart / development context where onsite power reliability, diesel logistics, capital allocation and contract structure can directly affect project execution. Public NT EPA material described the original Fountain Head proposal as recommencing the Fountain Head open pit, approximately 170 km south of Darwin, with a 3-year life of mine. [1]

The public greenhouse gas assessment described onsite diesel generation as the base power solution. The assessment refers to an onsite diesel-fired power station comprising three 1.5 MW units, total peak power of 3.9 MW, and 7,500 kL/year of diesel during operations. [2] For this screening model, the annual energy demand is reconstructed as approximately 29,959 MWh/year, consistent with a high utilisation remote mine load profile and the diesel consumption envelope used in the screening model.

The energy decision is therefore a commercial structuring question, not only a technical sizing question. The mine owner needs to understand whether a solar, battery and diesel hybrid system should be self-funded, funded by an EPC/vendor package, or structured as a third-party BOO / SPV / EaaS arrangement. The answer depends on mine life, load certainty, diesel price, residual value, debt capacity and the allocation of diesel generator costs.

2. The post-merger story: from standalone mine to expanded hub

The original Fountain Head framing was a short-life standalone mine. That matters because a short contract horizon weakens third-party financing. PV, BESS, diesel generation and integration infrastructure have useful lives beyond a 3-year mine plan, but the SPV can only finance against contracted cash flow. A short contract therefore pushes value into residual / buy-out payments, which can damage the owner's lifecycle savings.

The post-merger ownership context changes the discussion. Patronus reported that its merger with PNX Metals was implemented on 11 September 2024, strategically integrating PNX's Northern Territory gold, base metals and uranium projects into Patronus. [4] Patronus also describes its Pine Creek holdings as an under-explored district strategy and states that existing Pine Creek resources at Fountain Head and Mt Porter total 0.3 Moz gold. [3]

The broader asset context supports a longer-life screening case, but it does not prove it. This report uses a 7-year expanded hub assumption for S3, S4 and S5 to test how energy financeability changes if Fountain Head is treated as part of a longer regional operating platform. This is a modelling assumption for owner discussion. It must be reconciled to the owner's latest mine plan, processing strategy, drilling results, ore schedule and approval pathway before being used for a commercial commitment.

Regulatory progress also supports further investigation. Patronus announced in November 2024 that the NT Government had approved the amended Mining Management Plan for Fountain Head, noting that environmental approval had been granted under the NT Environment Protection Act 2019 on 17 February 2023 and that the approval represented a key milestone toward development. [5]

3. What the model tests

The model is a financeability screen, not a final engineering design. It tests whether different hybrid energy structures can simultaneously satisfy the mine owner, the SPV investor and a simplified lender DSCR threshold. It includes PV, BESS, diesel generation, diesel fuel, fixed and variable O&M, tariff revenue, debt sizing, required buy-out and customer true saving.

A key modelling choice is the SPV contract boundary. The diesel generator and residual diesel cost are included in the SPV boundary. This matters because a hybrid power provider cannot be evaluated only on PV and BESS CAPEX if it is expected to supply reliable mine power. If the owner wants a different allocation - for example owner-owned diesel generators with only PV/BESS supplied by the SPV - the model should be re-run as a separate commercial structure.

The model also separates headline tariff saving from true lifecycle saving. S3, S4 and S5 use the same customer tariff discount relative to the diesel-only baseline, so the owner's annual headline saving looks similar across those cases. The real difference is behind the tariff: renewable penetration changes SPV diesel fuel exposure, system CAPEX, required buy-out and therefore customer true saving after buy-out.

4. Scenario comparison

Scenario	Demand horizon	Renewable share	Financeability reading	Required buy-out	Customer true saving	Owner interpretation
S1 - Short-life public-data rebuild	3-4 years	c. 28%	Pass DSCR only with very low debt; fails owner economics	\$27.3m	-\$20.1m	Baseline sanity check; not a strong SPV case
S3 - Expanded hub, same sizing	7 years	c. 28%	Pass DSCR after commercial repair; owner economics still weak	\$14.5m	-\$5.36m	Shows longer life helps but sizing remains insufficient
S4 - 60% renewable hybrid	7 years	60%	Pass DSCR and positive customer true saving	\$4.49m	+\$4.63m	Current preferred screening case
S5 - 100% annual renewable screen	7 years	100%	Pass DSCR after re-gearing; fails owner economics	\$24.51m	-\$15.39m	Upper-bound decarbonisation screen only

Table 2. Summary of current screening results. Values are indicative and rounded.

5. The story behind each scenario

5.1 S1 - why the original short-life case does not work as a standalone SPV

S1 is useful as a baseline, but it should not be sold as a bankable third-party energy solution. The short mine-life window is too narrow to recover the integrated cost of PV, BESS, diesel generation and energy system integration. Even if the case can be forced to a 1.30x DSCR by using very low debt, the equity capital still needs a large residual / buy-out to reach its target return. That buy-out is economically unattractive for the owner.

The owner implication is practical: if Fountain Head were only a short standalone restart, the better path may be owner-funded diesel optimisation or a limited EPC package, not a full BOO / SPV energy-as-a-service structure.

5.2 S3 - longer life fixes the bankability problem, but not the owner economics

S3 tests the impact of treating Fountain Head as part of a 7-year expanded hub. The longer contract horizon materially improves debt capacity. However, because renewable penetration remains low, the SPV still carries substantial residual diesel fuel cost. That leaves the required buy-out too high relative to the owner's lifecycle savings.

The S3 lesson is important: mine-life extension is necessary for financeability, but not sufficient. The energy system also needs to be re-sized so that diesel exposure is materially reduced without oversizing the BESS.

5.3 S4 - why the 60% renewable case is the preferred screening case

S4 is the first case where the owner and the SPV both have a credible story. A 60% renewable hybrid system is large enough to materially reduce diesel fuel exposure, but not so large that BESS CAPEX overwhelms the economics. The result is lower required buy-out, stronger debt capacity and positive customer true saving after buy-out.

For the owner, S4 is easier to defend than both extremes. It does not depend on a 100% renewable claim that still needs detailed dispatch validation, and it does not leave the owner with a large residual payment that wipes out headline tariff savings. It provides a clearer basis for hourly modelling, EPC pricing and lender conversation.

5.4 S5 - why 100% renewable is not the commercial base case yet

S5 is valuable, but as a boundary case. It shows what happens if the project is pushed toward 100% annual renewable energy. The case can be re-gearred to pass 1.30x DSCR, but this only means debt has been reduced to fit the cash flow. It does not mean the owner economics are attractive.

The problem is the required buy-out. The larger PV and BESS system increases total investment cost and residual value. Even though diesel fuel exposure is greatly reduced, the owner's lifecycle savings are not enough to absorb the buy-out. Therefore S5 should be described as an upper-bound decarbonisation screen, not as the preferred commercial pathway.

6. Why headline annual saving is not enough

The owner should not decide based only on tariff discount. S3, S4 and S5 are all structured around the same diesel-only baseline and the same indicative customer discount. That means the annual customer saving looks similar. But this is not the true economic comparison.

The correct owner-side metric is customer true saving after buy-out. This metric compares the owner's contract-period gross saving with the residual / buy-out required to support SPV economics. It shows why S4 is attractive while S3 and S5 are not. S3 is still too diesel-heavy; S5 is too capital-heavy; S4 sits in the practical middle.

7. Recommended owner pathway

Next step	Purpose
Step 1 - Confirm the demand horizon	Validate whether a 7-year expanded hub energy demand assumption is realistic, using the latest mine plan, ore schedule, processing strategy and regional resource plan.
Step 2 - Validate S4 through hourly dispatch	Use hourly solar resource, mine load, BESS state of charge, wet-season performance, diesel backup, outage assumptions and reserve margins to test reliability.
Step 3 - Re-price with vendor and EPC inputs	Replace high-level CAPEX and OPEX assumptions with vendor quotations, EPC pricing, diesel generator configuration and site-specific logistics.
Step 4 - Agree the commercial boundary	Decide whether the owner or SPV owns diesel generators, who carries diesel fuel price risk, how buy-out is calculated, and how carbon benefits are allocated.
Step 5 - Convert to a bankable structure	Move from simplified DSCR screening to a lender-grade model with debt sculpting, DSRA/DSCA, tax losses, augmentation, covenants and downside cases.

Table 3. Recommended pathway from screening model to owner-side decision.

8. Implications for Fountain Head owner

The practical recommendation is to move forward with S4 as the base case for diligence. This does not require the owner to commit to a 60% renewable target today. It means the 60% hybrid case should be the reference design for the next round of hourly modelling, vendor pricing and commercial structuring.

The 100% renewable case should stay in the materials, but with the right label. It is useful for board and ESG discussion because it shows the capital and buy-out challenge associated with high renewable penetration. It should not be presented as the first commercial proposal unless the owner has a materially longer mine life, grant funding, a higher willingness to pay, or a strategic reason to own more of the energy infrastructure.

The biggest unresolved question is not the spreadsheet; it is the mine plan. If the owner can support a longer energy demand horizon through Fountain Head, Glencoe, Tally Ho, Mt Porter or other Pine Creek assets, the energy project becomes much more financeable. If the demand horizon falls back to a short standalone mine, the SPV structure becomes much harder to justify.

9. Disclaimer and scope limitation

This report has been prepared as an independent preliminary screening review for discussion purposes only. It is based on publicly available information, indicative assumptions, high-level desktop analysis and simplified financial modelling. It is intended to help frame the commercial, technical and financing questions that may be relevant to a hybrid solar, battery and diesel energy solution for the Fountain Head mining project.

This report does not constitute an engineering design, EPC proposal, definitive feasibility study, bankable feasibility study, investment recommendation, financing offer, valuation, legal advice, tax advice, accounting advice, environmental approval advice or formal technical due diligence. No party should rely on this report as the sole basis for an investment decision, financing decision, procurement decision, mine planning decision or contractual commitment.

The modelling outputs are indicative only. They depend on assumptions regarding mine life, annual energy demand, peak load, diesel cost, diesel generator configuration, PV sizing, BESS sizing, renewable penetration, capital costs, operating costs, financing terms, tariff structure, customer discount, residual value, buy-out amount, tax treatment and target investor return. These assumptions have not been verified against owner-provided operating data, vendor quotations, EPC pricing, lender term sheets, grid studies, permitting requirements or detailed mine plans.

The 60% renewable case and 100% renewable screening case are annual-energy screening cases. They do not prove operational reliability. Any renewable penetration target must be tested through hourly dispatch modelling, including solar resource, mine load profile, night-time demand, wet-season performance, battery state of charge, degradation, outage assumptions, diesel generator minimum load, reserve margin, ramping constraints and site-specific reliability requirements.

The discussion of SPV, BOO, EaaS, tariff, buy-out, residual value, debt sizing and DSCR is illustrative only. It does not represent a financing commitment, lender approval, investor approval or preferred commercial structure. Any project financing assessment would require detailed due diligence, lender engagement, legal documentation, tax structuring and credit assessment.

The discussion of carbon, emissions reduction or decarbonisation benefits is indicative only. Any carbon benefit, Safeguard Mechanism implication, ACCU / SMC value, emissions accounting outcome or customer ESG claim must be separately assessed against applicable Australian law, reporting standards, regulatory guidance and project-specific emissions data.

No confidential owner data has been used unless separately provided and expressly identified. Where publicly available information has been used, it has not been independently verified. The author accepts no responsibility for errors, omissions or changes in source information, project status, ownership, mine plan, commodity price, capital cost, financing market conditions, policy settings or regulatory requirements.

Appendix A - Model result snapshot

Item	Value	Comment
Annual energy demand	29,959 MWh/year	Screening model assumption based on public-data reconstruction
Peak power reference	3.9 MW	Public GHG assessment references 3 x 1.5 MW diesel units and 3.9 MW total peak power
S4 renewable share	60%	Current preferred screening case
S4 customer true saving after buy-out	+\$4.63m	Positive owner-side lifecycle economics
S5 selected gearing	c. 56.8%	Current-case back-solve to pass 1.30x DSCR
S5 customer true saving after buy-out	-\$15.39m	Fails owner economics despite DSCR pass
Target DSCR	1.30x	Simplified lender downside DSCR screen

Table A1. Rounded model result snapshot from the current v4.1 clean workbook.

Appendix B - Source notes

The public-source statements in this report rely on the following references. The financial model outputs are from the internal Fountain Head Two Scenario Financeability v4.1 clean workbook and remain indicative.

[1] NT EPA public register - Fountain Head Gold Project. <https://ntepa.nt.gov.au/your-business/public-registers/environmental-impact-assessments-register/completed-assessments/register/fountain-head-gold-project>

[2] Katestone / ERIAS, Fountain Head Gold Project Air Quality and Greenhouse Gas Assessment, 30 March 2021. https://ntepa.nt.gov.au/_resources/documents/eia/pnx-metals-limited/draft-environmental-impact-statement-eis/appendices/1-air-quality-and-greenhouse-gas-assessment.pdf

[3] Patronus Resources, Northern Territory project page. <https://www.patronusresources.com.au/northern-territory>

[4] Patronus Resources September 2024 Quarterly Activities Report, 23 October 2024. <https://announcements.asx.com.au/asxpdf/20241023/pdf/069g6qx86vg461.pdf>

[5] Patronus Resources, Key Mining Approval Received for Fountain Head Gold Project, 4 November 2024. <https://clients3.weblink.com.au/Clients/JuniorMining/headline.aspx?id=61235883>

Appendix C - Suggested owner discussion questions

- Does the owner support a 7-year energy demand horizon for Fountain Head / Pine Creek hub planning, or should the model retain a shorter contracted case?
- Which loads are firm, deferrable or seasonal, and what hourly load data can be provided for dispatch validation?
- Should diesel generators be owned by the owner or included inside the SPV boundary?
- Is the owner willing to accept a buy-out mechanism, and should it be fixed, formula-based, or optional?
- Who should retain carbon, Safeguard Mechanism or ESG benefits from diesel displacement?
- Is the owner seeking lowest delivered energy cost, maximum renewable share, lower upfront capital, or a bankable energy services structure?